

## DESIGNATIONS AND ABBREVIATIONS

|          |                                                                            |
|----------|----------------------------------------------------------------------------|
| AI       | Artificial Intelligence                                                    |
| ALO      | Attention–Lesion Overlap (explainability metric)                           |
| APTOS    | Asia Pacific Tele-Ophthalmology Society (2019 Blindness Detection dataset) |
| AUC      | Area Under the Curve                                                       |
| BYOL     | Bootstrap Your Own Latent (self-supervised learning method)                |
| CAM      | Class Activation Mapping                                                   |
| CLAHE    | Contrast-Limited Adaptive Histogram Equalization                           |
| CNN      | Convolutional Neural Network                                               |
| CNR      | Contrast-to-Noise Ratio                                                    |
| DDR      | Diabetic Retinopathy grading dataset (DDR)                                 |
| DINO     | self-Distillation with No labels (self-supervised learning method)         |
| DR       | Diabetic Retinopathy                                                       |
| ECE      | Expected Calibration Error                                                 |
| EHR      | Electronic Health Record                                                   |
| EMD      | Earth Mover's Distance                                                     |
| EyePACS  | Eye Picture Archive Communication System (diabetic-retinopathy dataset)    |
| FOV      | Field of View                                                              |
| GDPR     | General Data Protection Regulation                                         |
| Grad-CAM | Gradient-weighted Class Activation Mapping                                 |
| HIPAA    | Health Insurance Portability and Accountability Act                        |
| IDRiD    | Indian Diabetic Retinopathy Image Dataset                                  |
| IoU      | Intersection over Union                                                    |
| KL       | Kullback–Leibler divergence                                                |
| LAB      | CIELAB colour space                                                        |
| MLP      | Multilayer Perceptron                                                      |
| MoCo     | Momentum Contrast (self-supervised learning method)                        |
| NPV      | Negative Predictive Value                                                  |
| OD       | Optic Disc                                                                 |
| ODIR-5K  | Ocular Disease Intelligent Recognition dataset (5,000 patients)            |
| PACS     | Picture Archiving and Communication System                                 |
| PPV      | Positive Predictive Value                                                  |
| ReLU     | Rectified Linear Unit                                                      |
| RFMiD    | Retinal Fundus Multi-disease Image Dataset                                 |
| RIQA     | Retinal Image Quality Assessment                                           |
| ROC      | Receiver Operating Characteristic                                          |
| SimCLR   | Simple framework for Contrastive Learning of Representations               |
| SSIM     | Structural Similarity Index                                                |
| SSL      | Self-Supervised Learning                                                   |
| STARE    | Structured Analysis of the Retina (dataset)                                |
| UML      | Unified Modeling Language                                                  |

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| VRAM        | Video Random-Access Memory                                                                  |
| VVI         | Vessel Visibility Index                                                                     |
| WSL         | Windows Subsystem for Linux                                                                 |
| XAI         | Explainable Artificial Intelligence                                                         |
| D           | Field-of-view diameter, in pixels                                                           |
| CL          | Clip limit of the CLAHE transform                                                           |
| F1          | F1-score (harmonic mean of precision and recall)                                            |
| G           | Generalization ratio, $G = F1\_external / F1\_EyePACS$                                      |
| T           | Global threshold of the dual-constraint CLAHE (T/80 formulation)                            |
| $\alpha$    | Class-weight parameter of the Focal Loss (inverse-frequency)                                |
| $\gamma$    | Focusing parameter of the Focal Loss ( $\gamma = 2$ )                                       |
| $\kappa$    | Cohen's Kappa with quadratic weights                                                        |
| $\sigma$    | Standard deviation of the Gaussian kernel in flat-field correction, $\sigma = 0.07 \cdot D$ |
| $\Delta F1$ | Cross-dataset F1 drop, $\Delta F1 = F1\_EyePACS - F1\_external$                             |